

Dental caries

Community Lab 3

Third stage | college of Dentistry

Introduction

is defined as irreversible multi-factorial disease of microbial origin affecting the calcified tissues of the teeth in which the dietary carbohydrates are fermented by bacteria forming an acid that causes demineralization of the inorganic part and disintegration of the organic part of the tooth.

Composition of tooth

Enamel:

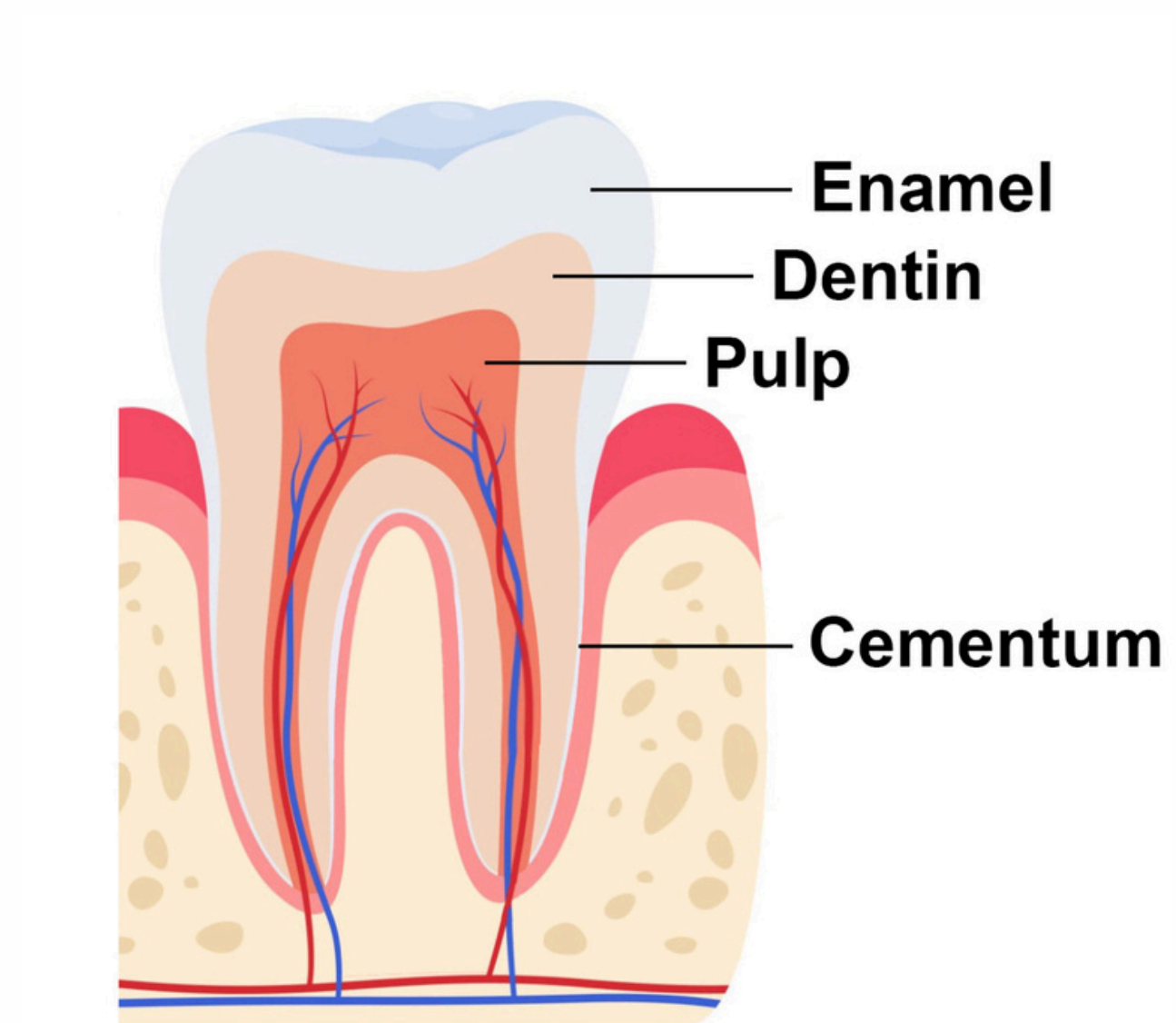
- Inorganic: 96%
- Organic + water: 4%

Dentin:

- Inorganic: 65%
- Organic matter + water: 35%

Cementum:

- Inorganic: 45-50%
- Organic + water: 50- 55%

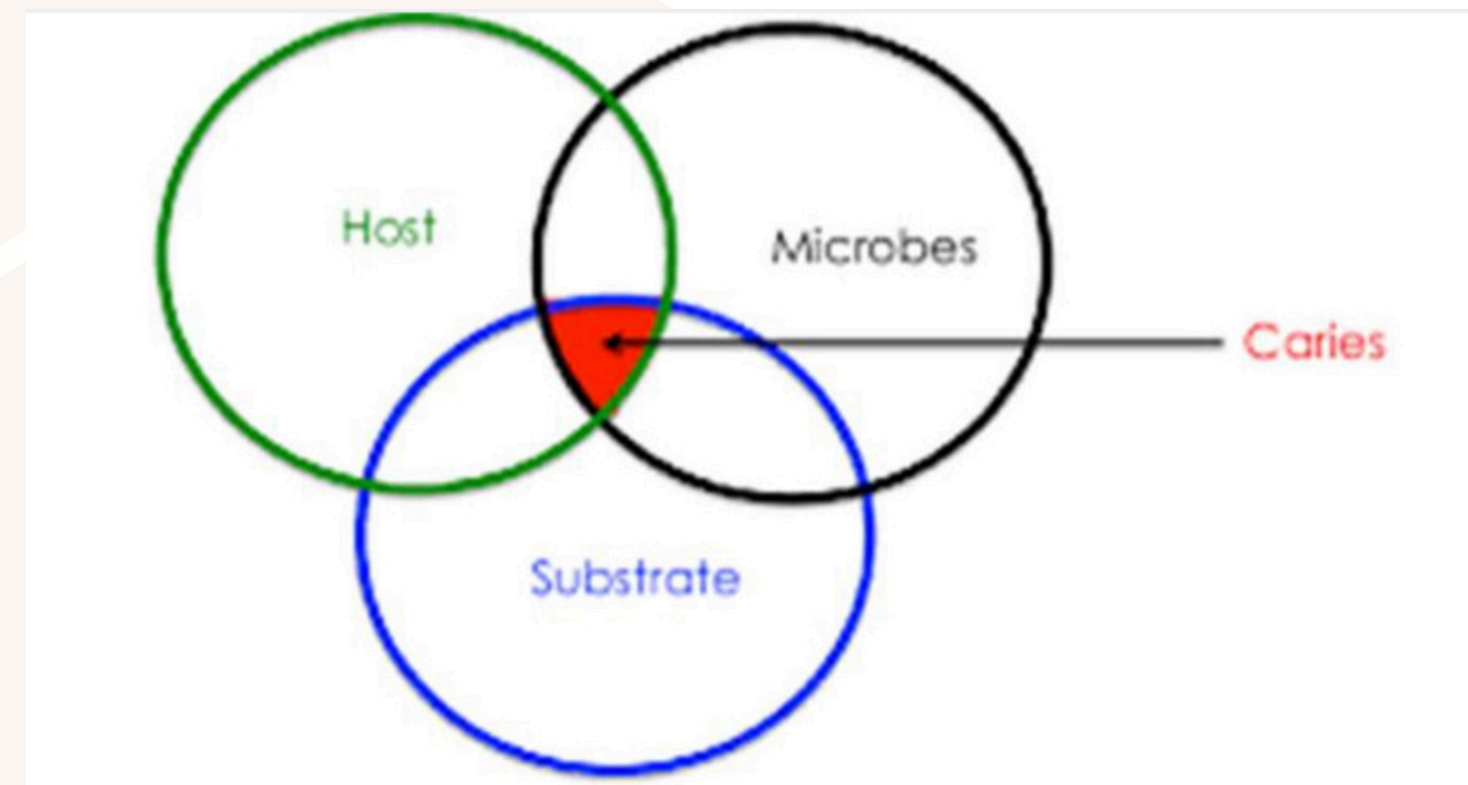


Dental caries

it is a multifactorial disease characterized by demineralization of the hard tissue of tooth .

Carious process is the result of an interaction of the following:

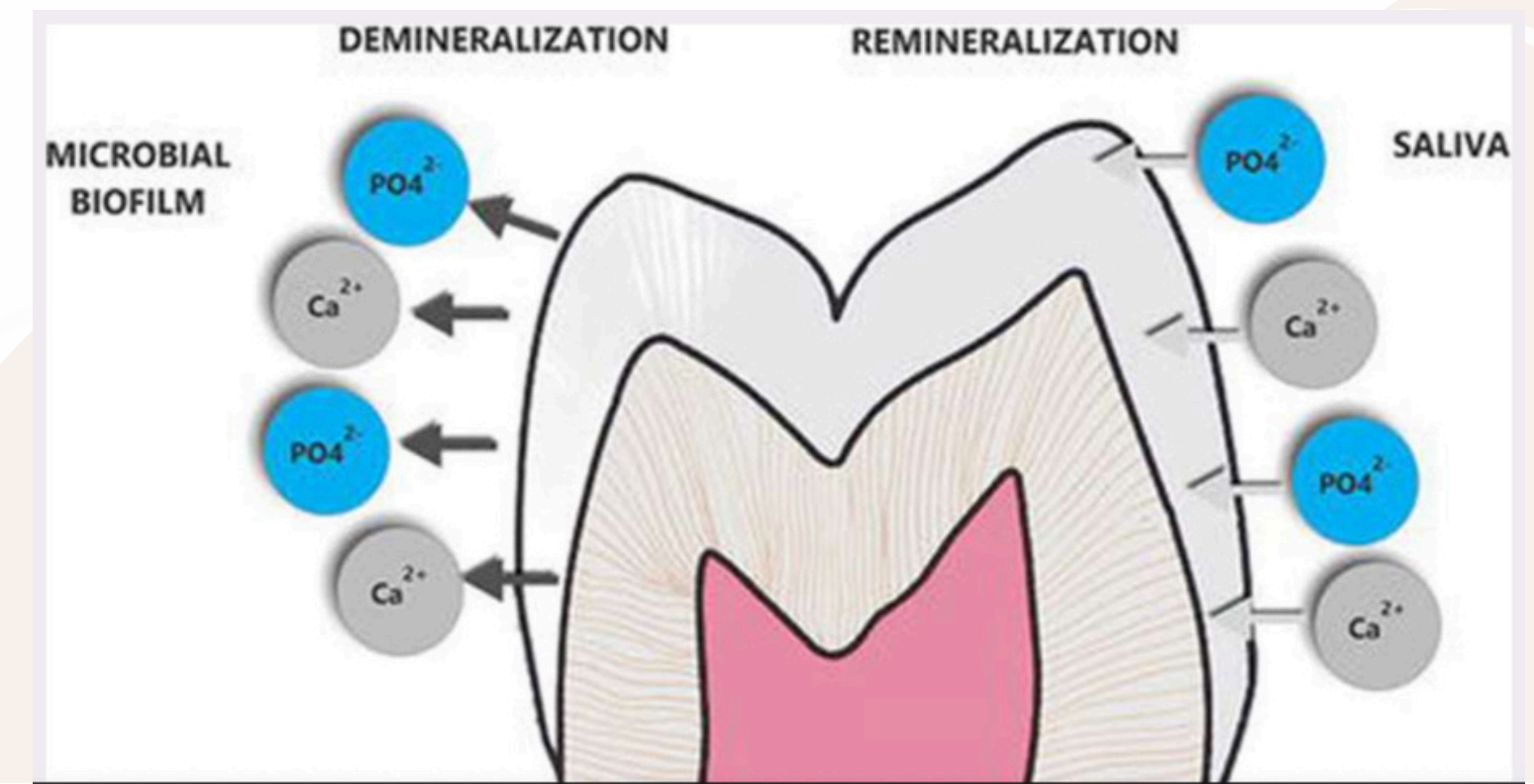
- 1_ Host.
- 2_ Plaque.
- 3_ Diet.
- 4_ Time.



During demineralisation, calcium and phosphorus ions are drawn out from tooth structure.

- The process of **remineralisation** involves the **deposition of minerals**, primarily **calcium** and **phosphate**, back into the enamel structure.

This can occur naturally through the presence of **saliva**, which **contains** **these essential minerals**



Classification of dental caries

- According to **number of surfaces**
- According to **Anatomic site**
- According to **Histology**
- According to **severity**
- According to **onset**
- According to the **rapidity of caries progression**
- According to **Chronology**
- According to **Activity**

1. According to number of surfaces involved:

- Simple: involving only one tooth surface
- Compound: involving two surfaces of tooth
- Complex: involves more than two surfaces of a tooth



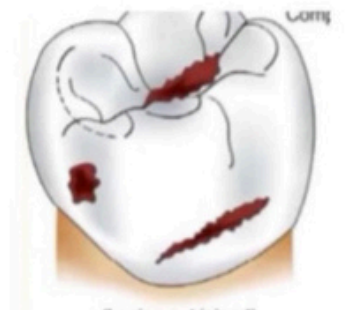
SIMPLE CARIES



COMPOUND
CARIES

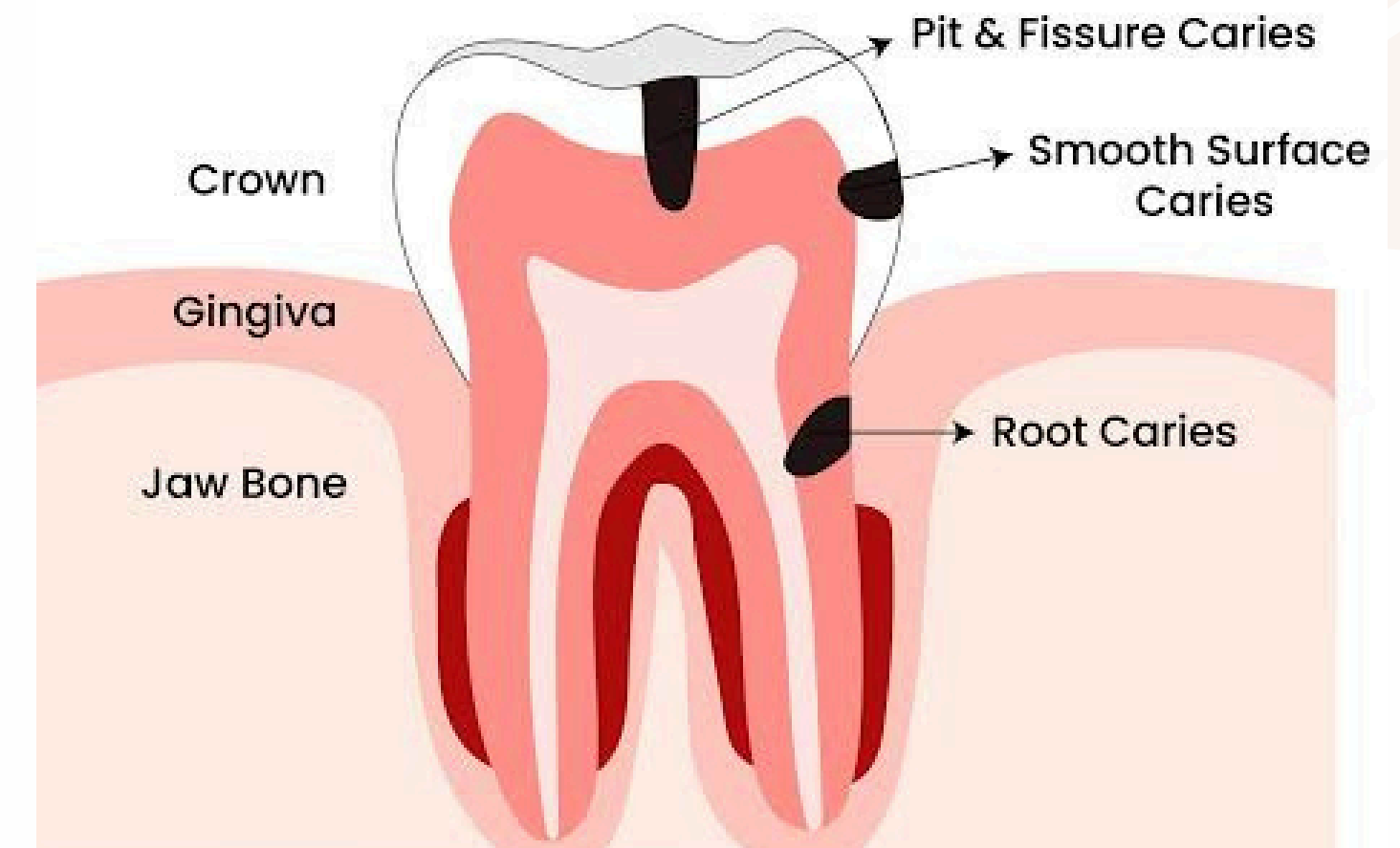


COMPLEX
CARIES



2. According to Anatomic site:

- Occlusal (pit & fissure) caries.
- Smooth surface caries (proximal & cervical caries)
- Root caries.

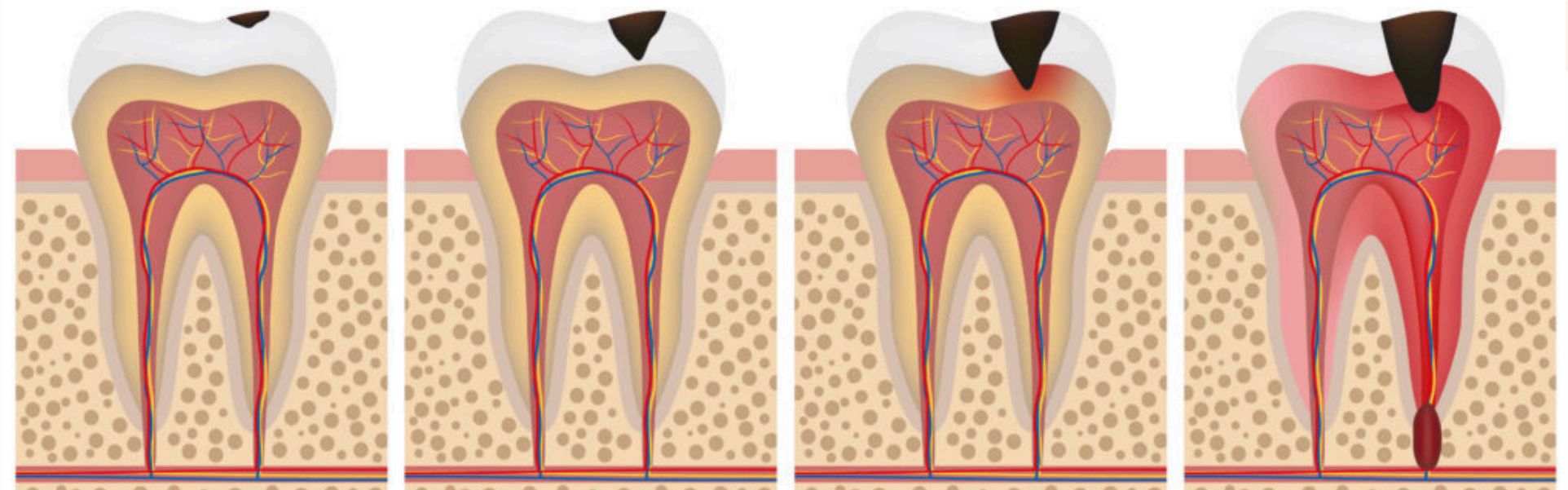


3. According to Histology:

- Enamel caries
- Dentinal caries
- Cemental caries

The stages of tooth decay

1. Decay in enamel 2. Advanced decay 3. Decay in dentin 4. Decay in pulp



4. According to severity:

- Incipient caries (Initial or primary): Carious lesion appears as a white opaque region (white spot lesion).

أحيانا تصير عند
أصحاب التقويم



4. According to severity:

- Occult (hidden) caries: not detected clinically only by radiographs.

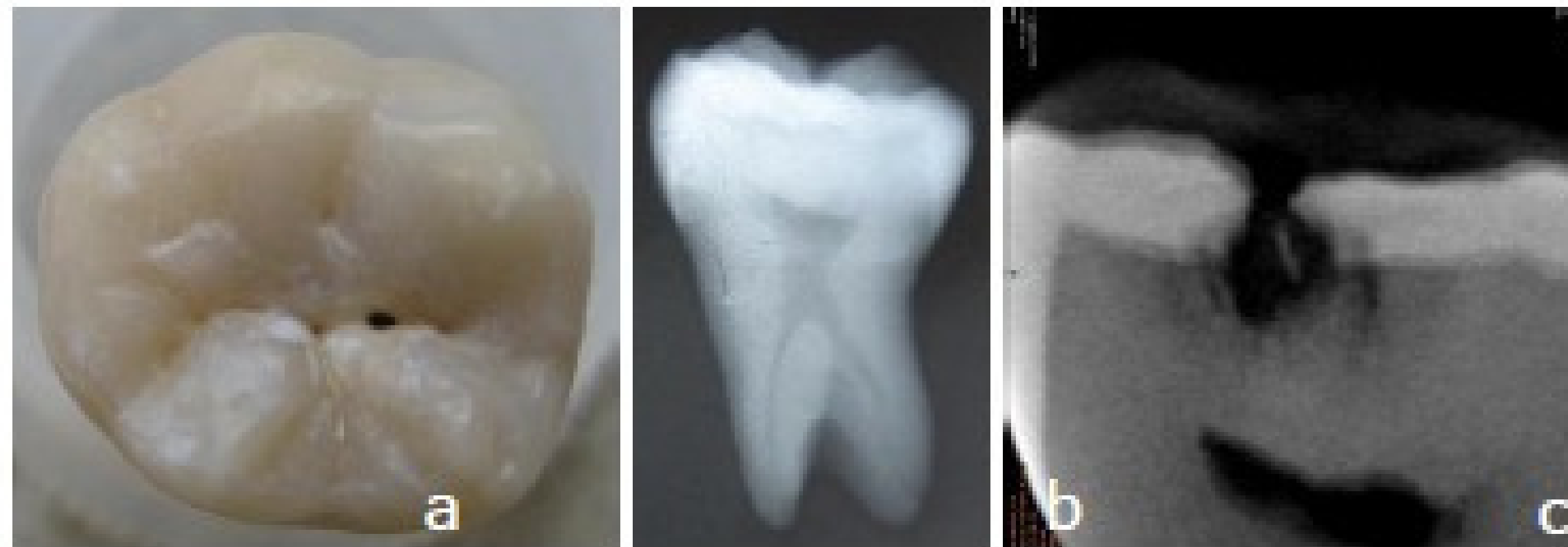


Figure 2. Tooth E: (a) photo of occlusal surface, (b) radiographic image, and (c) micro-CT scan image.

4. According to severity:

- Cavitation caries: The enamel surface is broken (not intact) and the lesion has advanced into enamel/dentin. No remineralization is possible at this stage.



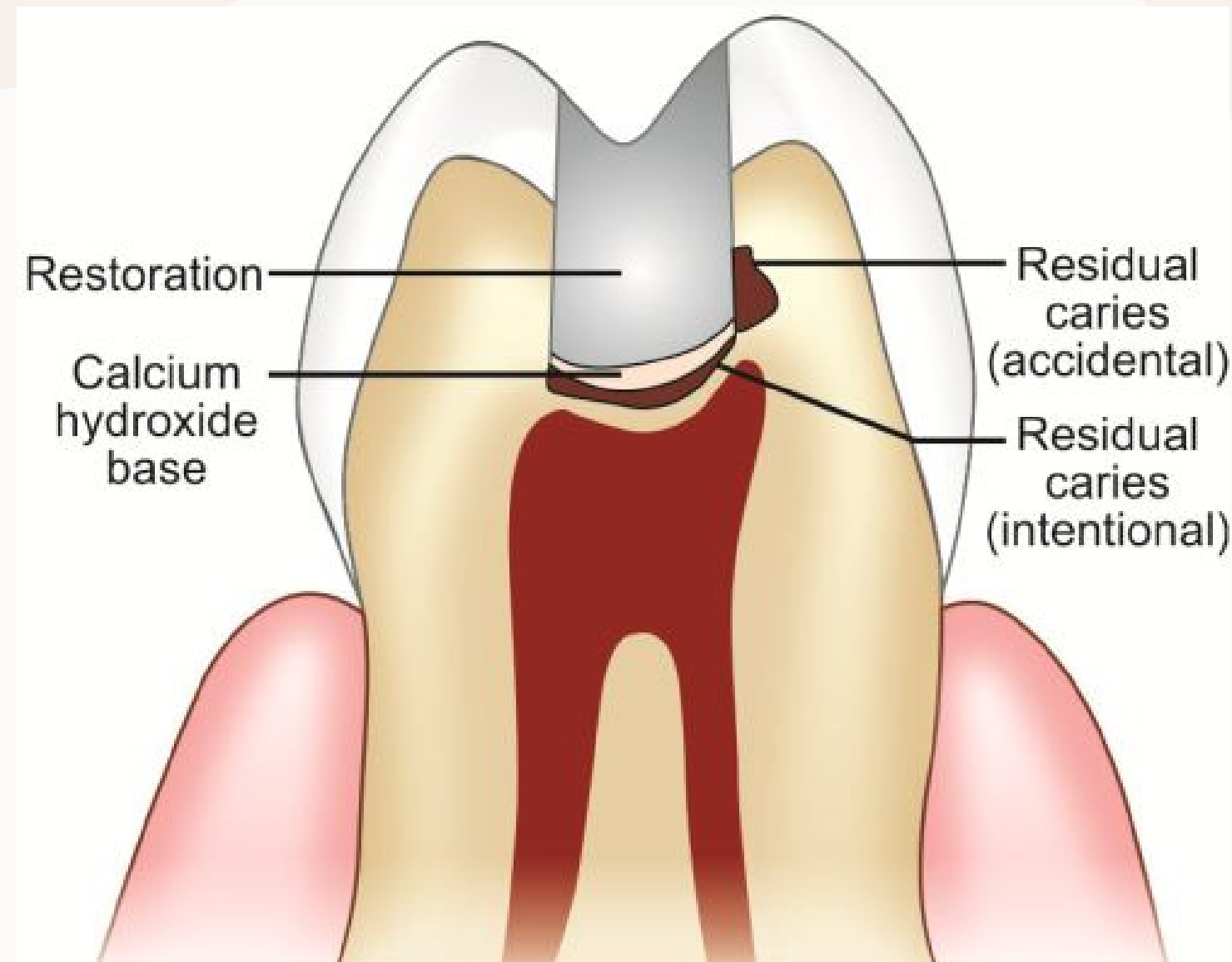
5. According to onset:

- Primary caries: is the original carious lesion of the tooth
- Secondary (recurrent) caries: Occurs at the interface of tooth and restorative material.



5. According to onset:

- Residual caries: is demineralized tissue left in place before a filling is placed.



6. According to the rapidity of caries progression:

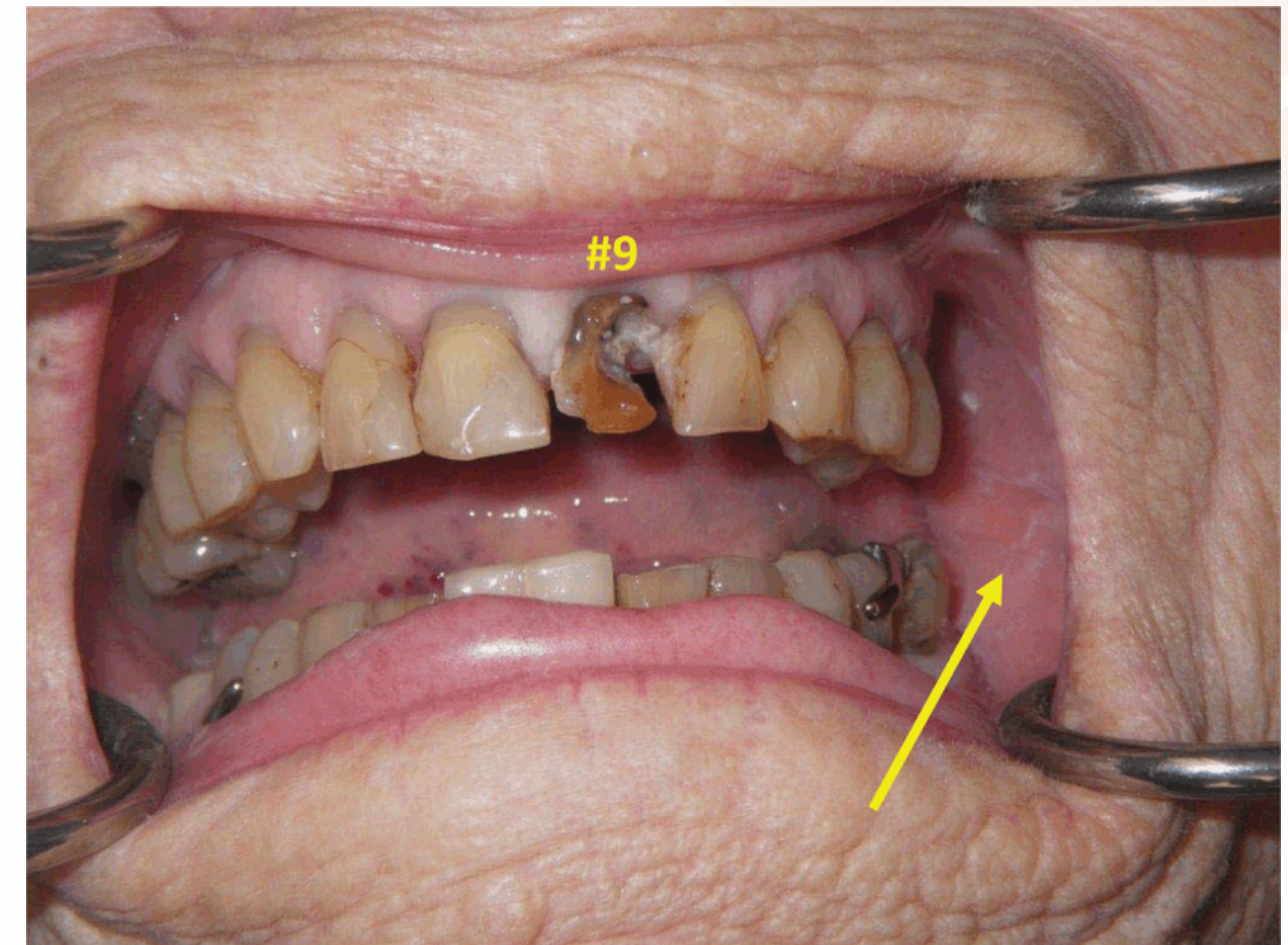
- Acute (rampant) caries
- Chronic caries



Nursing caries: – Acute caries occur in the primary teeth, 1 to 3 years old. Attributed to the practice of putting the infant to bed with a bottle of sweetened drink. More prevalent in low socioeconomic status population, where infants are being cared by little educated mothers. Prevention based on education of parents.

7. According to Chronology:

- Early childhood caries
- Adolescent caries
- Senile caries



8. According to Activity:

- Active carious lesion: progressive lesion.
- Arrested caries (inactive carious lesion): It is that caries which becomes static or stationary and does not progress any further and could occur in both deciduous and permanent dentition.

Clinical Diagnosis of Dental Caries

- Good lighting.
- Dry and clean teeth.
- Good vision and Bitewing radiographs.

Diagnosis of caries by one or all of the following:

- 1_ Visual inspection: Visual changes in tooth surface texture or color, it is subjective as assessment of features such as color and texture are qualitative in nature which provide some information on the severity of the disease but fall short of true quantification and their ability to detect early, non cavitated lesions restricted to enamel is poor.
- 2_ Tactile sensation: blunt Probe to detect any cavitation.
- 3_ Radiographs.
- 4_ Transillumination: in which light source directed through the tooth from the lingual side.

Caries susceptibility of individual teeth

- Upper and lower first molar → 95%
- Upper and lower second molar → 75%
- Upper second bicuspid → 45%
- Upper first bicuspid → 35%
- Lower second bicuspid → 35%
- Upper central and lateral incisor → 30%
- Upper cuspids and lower first bicuspid → 10%
- Lower central and lateral incisor → 3%
- Lower cuspids → 3%



Caries susceptibility of individual tooth surface:
occlusal > mesial, distal > buccal > lingual

Maxillary teeth more susceptible than mandibular teeth:
relate to gravity and saliva, with its buffering action, would
tends to drain from upper teeth and collect around lower teeth



Thank you!